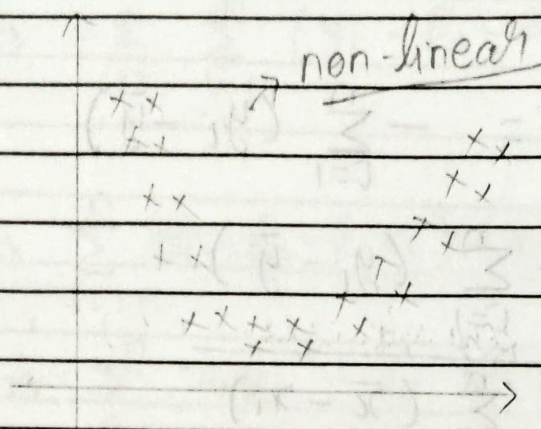


Polynomial Regression



$$h_0(x) = \theta_0 + \theta_1 x$$

↓

Simple Linear
Regression

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots$$

↳ multiple Linear regression

Simple Polynomial Regression → $\begin{cases} 1 \text{ I/P and} \\ 1 \text{ O/P feature} \end{cases}$

(i) $\text{deg} = 0$

$$h_0(x) = \beta_0 x^0 \rightarrow \text{constant value}$$

(ii) $\text{deg} = 1$

$$h_0(x) = \beta_0 x^0 + \beta_1 x^1 \rightarrow \text{Simple Linear Reg.}$$

(iii) $\text{deg} = 2$

$$h_0(x) = \beta_0 x^0 + \beta_1 x^1 + \beta_2 x^2 \rightarrow \text{Simple Poly}$$

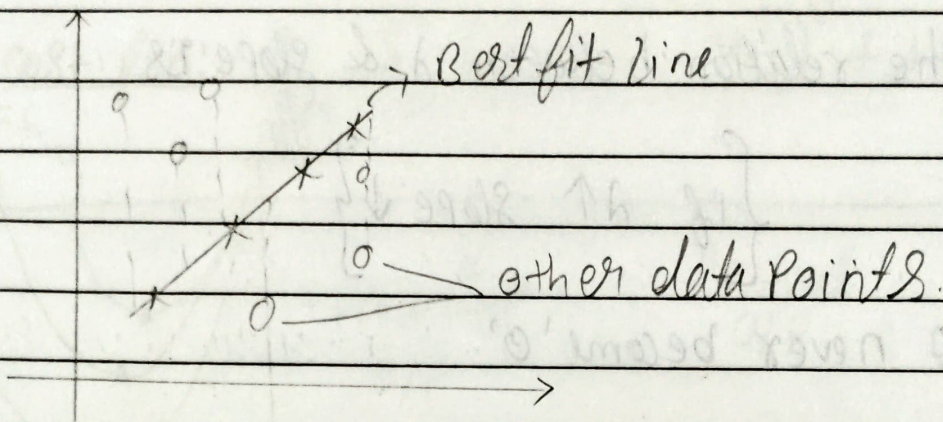
Ridge Regression.



(L-2 Regularization)



(Reduce the overfitting)



→ Overfitting → Training acc ↑ { Low Bias
 ↓
 → Testing acc ↓ { High variance.

Reduce this we
 use the Ridge Regression.

So our model is over fitted because
 of the cost function becomes "0".

$$\text{Cost fun} = \frac{1}{2m} \left(\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \right)$$

↓

For Linear Regression.

$$\text{cost fun} \Rightarrow \frac{1}{2m} \sum_{i=1}^m (h_0(x)^i - y^i)^2 + \lambda \sum_{i=1}^n (\text{slope})^2$$

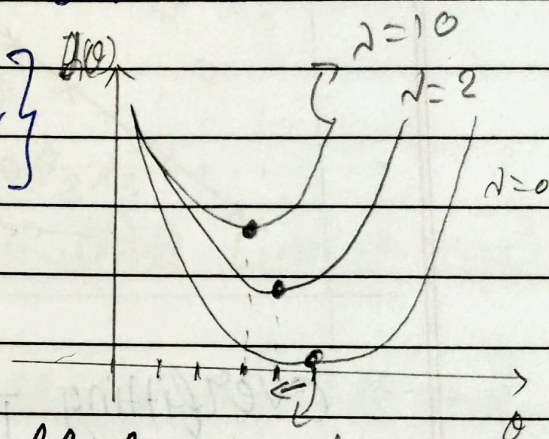
Hyperparameter

→ for Ridge Regression.

The relation between λ & slope is

{ if $\lambda \uparrow$ slope $\downarrow$ }

→ θ never become '0'.



→ Ridge Regression in multiple-Linear Regression: global minima

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_3^2$$

⇒ So in this case all the θ (slope) values are decreasing by a factor so that it is adjust the line for better performance.

Lasso Regression

(L-1 Regularization)

(feature selection)

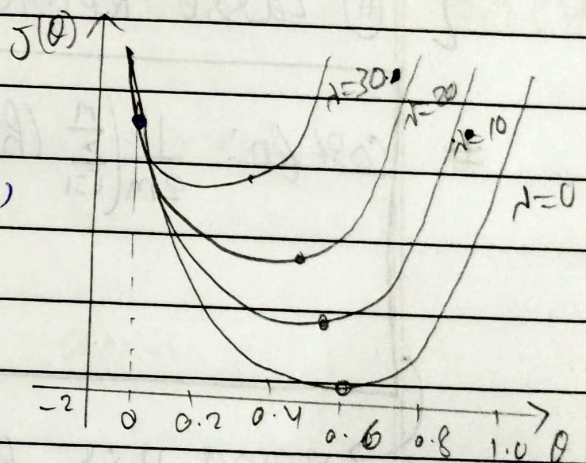
$$\text{cost fun: } \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x)^i - y^i)^2 + \lambda \sum_{i=1}^n |\text{slope}|$$

predicted
O/P

actual
O/P

magnitude

⇒ at one point of time the λ value becomes '0' so that we have to remove that feature from our model.



Let's assume →

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

$$h_{\theta}(x) = 0.52 + 0.65x_1 + 0.72x_2 + 0.34x_3 + 0.12x_4$$

↳ applying Lasso Regression.

$$h_{\theta}(x) = 0.52 + 0.52x_1 + 0.63x_2 + 0.25x_3 + 0x_4$$

Remove this.

The λ value which becomes zero is not co-related to the features so we have to remove it.

ElasticNet Regression

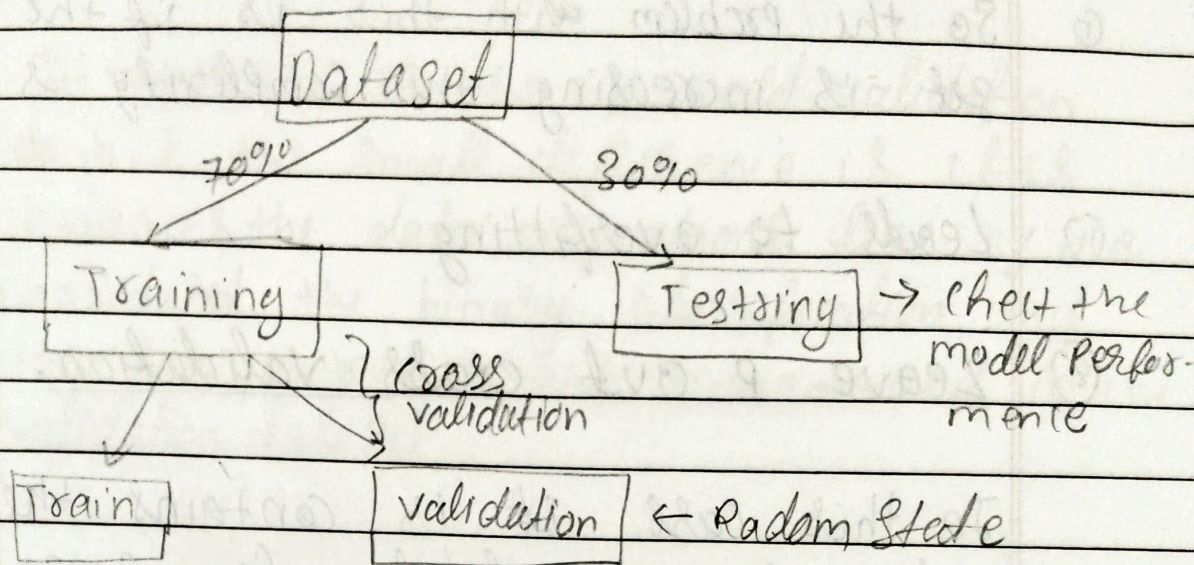
It consist the Both regression type

- ① Ridge Regression $\rightarrow$ Reducing overfitting
- ② Lasso Regression $\rightarrow$ Feature selection.

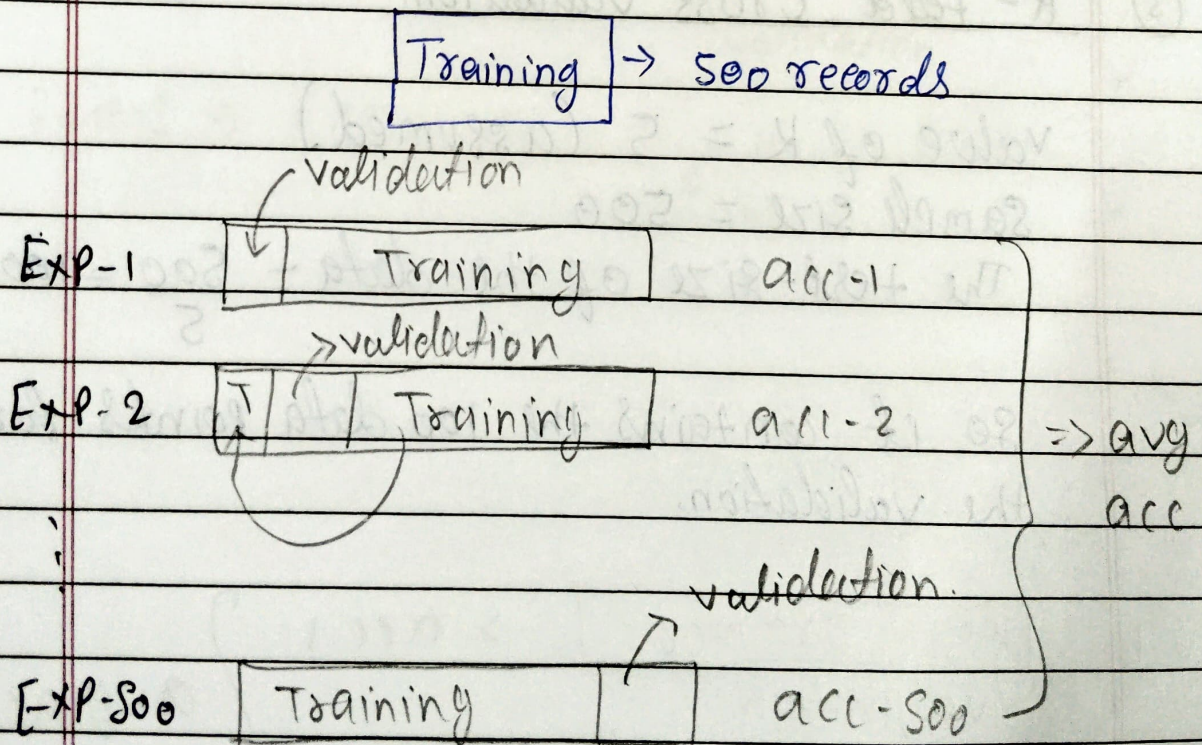
$$\# \text{ cost fn: } \frac{1}{2m} \left(\sum_{i=1}^m (h_{\theta}(x)^i - y^i)^2 \right) + \lambda \sum_{i=1}^n (\text{slope})^2 + \lambda \sum_{i=1}^n |\text{slope}|$$

$\rightarrow$ using this for hyper-parameter tuning for the Linear Regression.

Concept of Cross validation.



① Leave one out cross validation (LOOCV)



- ① So the problem with this is if the data-set is increasing the complexity is inc↑
- ② Leads to overfitting.
- ③ Leave P out cross validation.

In this case it contains the P value for the validation. like $P=20, 30, \dots$ and others all same process.

③ K-Fold cross validation:

value of $K = 5$ (assumed)

sample size = 500

The test-size of the data = $\frac{500}{5} = 100$

So it contains the 100 data points for the validation.

CV 1	100	Training	→ acc 1	} avg acc of all the CV
	validation			
CV 2	100	Training	→ acc 2	
		validation		
CV 3		Training	→ acc 3	

④ Stratified K Fold cross validation:

It works as the K-fold validation but the small difference is it balances the data in ratios like if we considered the binary classification and the 60:40 ratio are maintained in the validation dataset.

